Issue	**01**
Factory ID	**25796**
Date of Issue	**28-Jul-2025**

Ebrahim Textile Mills

PLOT HX2,ZONE 2LANDHI INDUSTRIAL AREA KARACHI
Karachi
Sindh Pakistan

Latitude: [24.840655], longitude: [67.239885]
Date of inspection: 13-May-2025

Fire Safety Inspection Report

Observations & Actions



Authors	**Waqar Sami**
Reviewed by	**Usama Zia**
Approved by	**Paul Rigby**

Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 13-05-2025 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

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1. Executive Summary

On 13-May-2025, Waqar Sami and Dr. Muhammad Imran on behalf of the Pakistan Accord carried out an initial fire inspection for Ebrahim Textile Mills, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Syed Hurr (Sr Manager Compliance), Mr. Aqeel Mohsin (G.M Engineering), Mr. Jamil Hussain (Electrical Manager), Mr Khobaib Javaid (Manager Civil) and other staff members of Ebrahim Textile Mills (Pvt) Ltd. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection.
The factory operates with one general shift and three shifts for processing only. Approximately 1380 workers (peak occupants) operate in the single shift of the factory. Factory employs 1380 workers out of which approximately 2% employees are female.
The factory comprises 6 buildings owned by Ebrahim Textile Mills (Pvt) Ltd. During the inspection, the buildings highlighted in Figure 1.1 were surveyed, with the site boundary indicated in red: The building highlighted with green in figure 1.1 was not inspected during the initial inspection. This building shall be included in future follow-up inspections.

- **Building 1-** a 1-storey and 2-storeys (Ground only and Ground + 1) multiple structures consists of reinforced concrete (RC) moment resisting frame (MRF) and Steel MRF with masonry infill walls. The building is used for, empty (1st floor) and admin HR, godowns, chemical godown, warehouse, workshops, carton godowns (ground floor). The building was constructed in 1991.
- **Building 2-** a 1-stoery and 2-storeys (Ground only and Ground + 1) multiple structures consists of reinforced concrete (RC) moment resisting frame and Precast Girder slab resting on RC moment resisting frame with masonry infill walls. The building is used for, marketing office (1st floor) and digital printing, embroidery, main store, local folding, masjid, workshops (ground floor). The building was constructed in 1991.
- **Building 3-** a 1-storey (Ground only) multiple structures consist of reinforced concrete (RC) moment resisting frame and Precast Girder slab resting on RC moment resisting frame with masonry infill walls. The building is used for greige department and engineering offices (ground floor). The building was constructed in 1991.
- **Building 4-** is a 1-storey (Ground only) reinforced concrete (RC) moment resisting frame with inverted beams at roof and masonry infill walls. The building is used for stitching and warehouse (ground floor). The year of construction of the building is 1991.
- **Building 5-** is consists of 5 different structures which are interconnected at ground level such that you can move freely between them. The building was constructed in 1991 and a part of the building extended horizontally in 2013 and vertically in 2022.
- **Structure 1** – 1-storey (Ground only) reinforced concrete (RC) half shell structure, used for processing, folding, storage (ground floor).
- **Structure 2** – 1-storey (Ground only) Steel Portal frame, used for washing and storage (ground floor).
- **Structure 3** – 1-storey (Ground only) RC MRF structure and Precast slab resting on RC MRF, used for bleaching and labs (ground floor).
- **Structure 4** - 5-storey (G+M+3) RC moment frame building with masonry infill walls and roof is vertically extended with Steel portal frame, is used for cutting (roof), stitching (3rd and 2nd floor), engraving (1st floor), fgs storage (mezzanine) and processing and folding (ground floor). Mezzanine floor is a complete floor and have an area equal to other floors therefore, Mezzanine floor is considered as a complete floor from fire

point of view.

- **Structure 5** - 2-storeys (Ground +1) RC MRF and Steel portal frame structure is used for quilting (1st floor) and cutting (ground floor).
- **Building 6-** is consist of 2 different structures. The year of construction of the building is 1991 with horizontally extended with steel shed (Steel Portal Frame) in 2018 and 2021.
- **Structure 1**- a 1-storey (Ground only) RC moment resisting frame with masonry infill walls. The building is used for local processing (ground floor).
- **Structure 2**- a 1-storey (Ground only) Steel Portal frame structure. The building is used for local processing (ground floor).

We were presented with site layout, as-built Architectural drawings of all the buildings, civil denfense certificate, Emergency action plan, fire drill record, testing & maintenance record of FLS system. Further details on documentation are provided in Table 1.
A summary of the main concerns identified during the inspection includes (non-exhaustive):
Lockable doors installed that could prevent safe evacuation of staff.

- Lockable doors installed that could prevent safe evacuation of staff.
- Storage and Production areas not separated with fire rated construction.
- Stairs are not protected from fire and smoke spread.
- Fire pump is not listed and have negative suction.
- Sprinklers system is not available in the building.
- Standpipes outlets are missing in the stairs.
- Protected stairs discharge internally to the building.
- External discharge route is passing through high risk area.

There is natural gas line passing in the facility and have no automatic shut off valve in case of emergency, thus, it is recommended to have install automatic shut off valve in gas line at the starting point of gas inlet connection to the factory.

We see no reason to suspend operations in the facility due to these concerns. Actions with associated priorities and timeframes are given in the detailed section of this report. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view) Red dashed line indicates site boundary, and Yellow dashed line indicates building boundaries, highlighted (blue) area indicates buildings inspected and highlighted (green) will inspect in future follow-up inspection

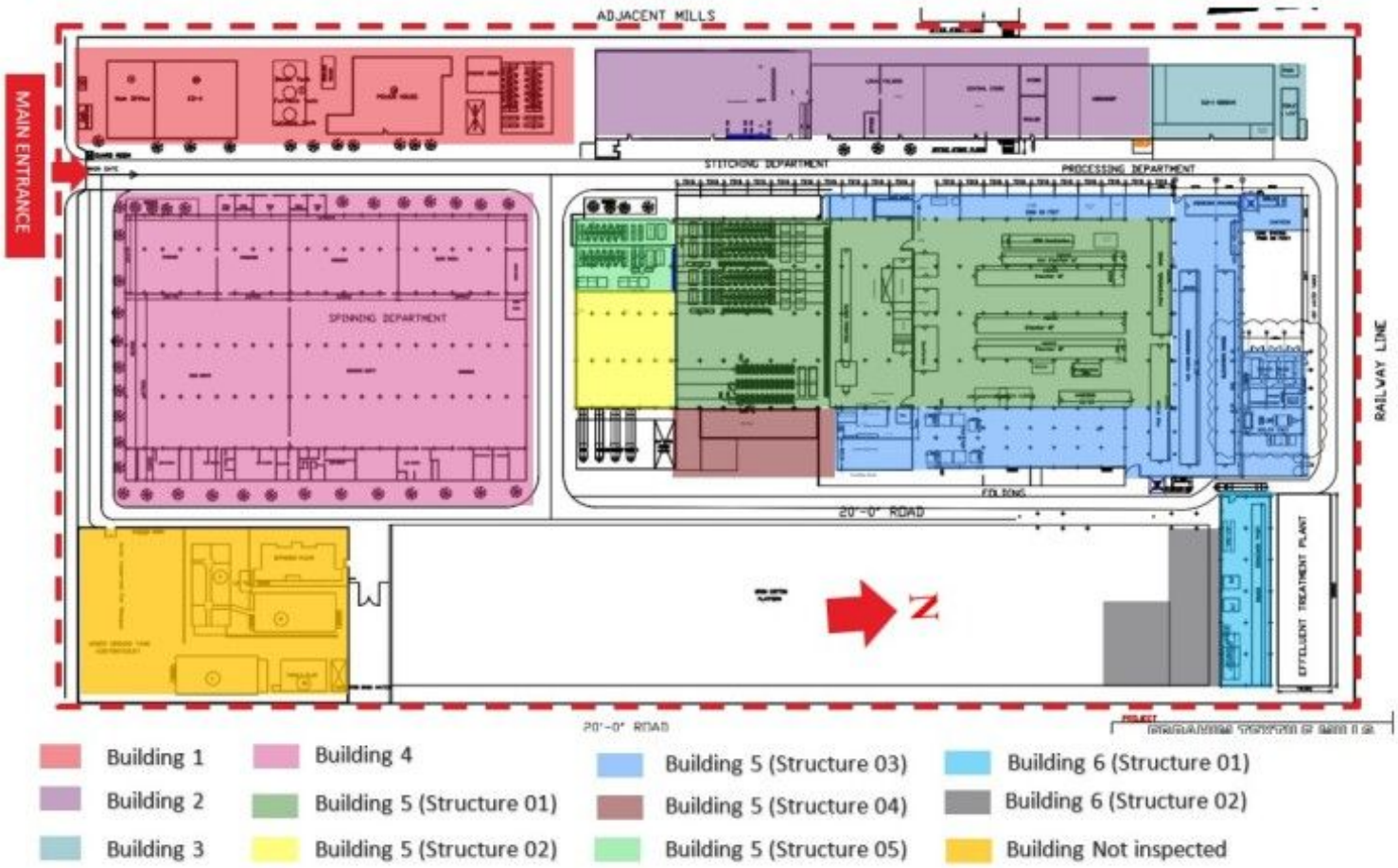


Figure 1.2. Site layout. Red dashed line indicates site boundary

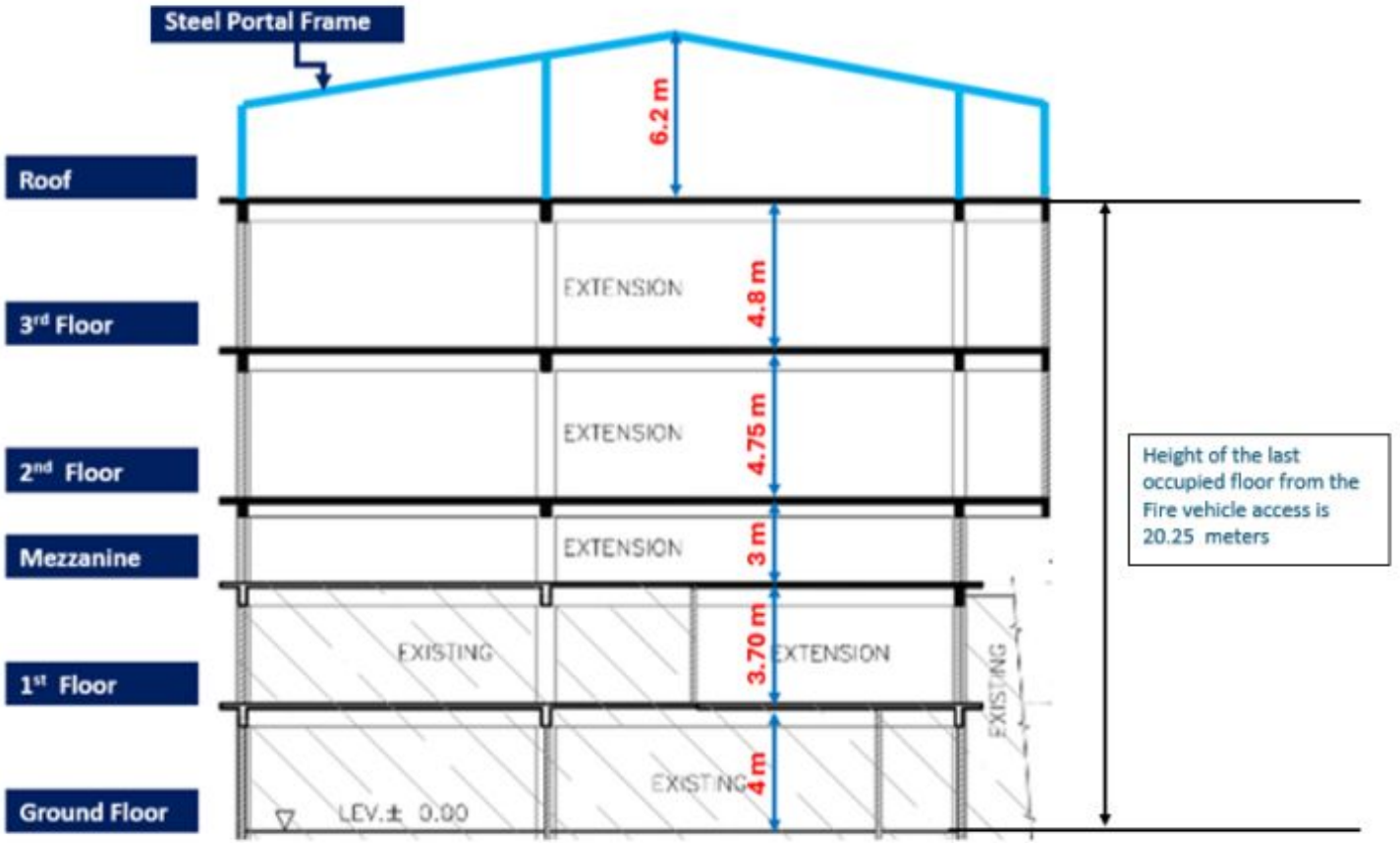


Figure 1.3. Typical section of building 5 (Structure 4)

3. Building Description

The factory comprises 6 buildings owned by Ebrahim Textile Mills (Pvt) Ltd. During the inspection, the buildings highlighted in Figure 1.1 were surveyed, with the site boundary indicated in red: The building highlighted with green in figure 1.1 was not inspected during the initial inspection. This building shall be included in future follow-up inspections.

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3.1 Building Means of Egress Details

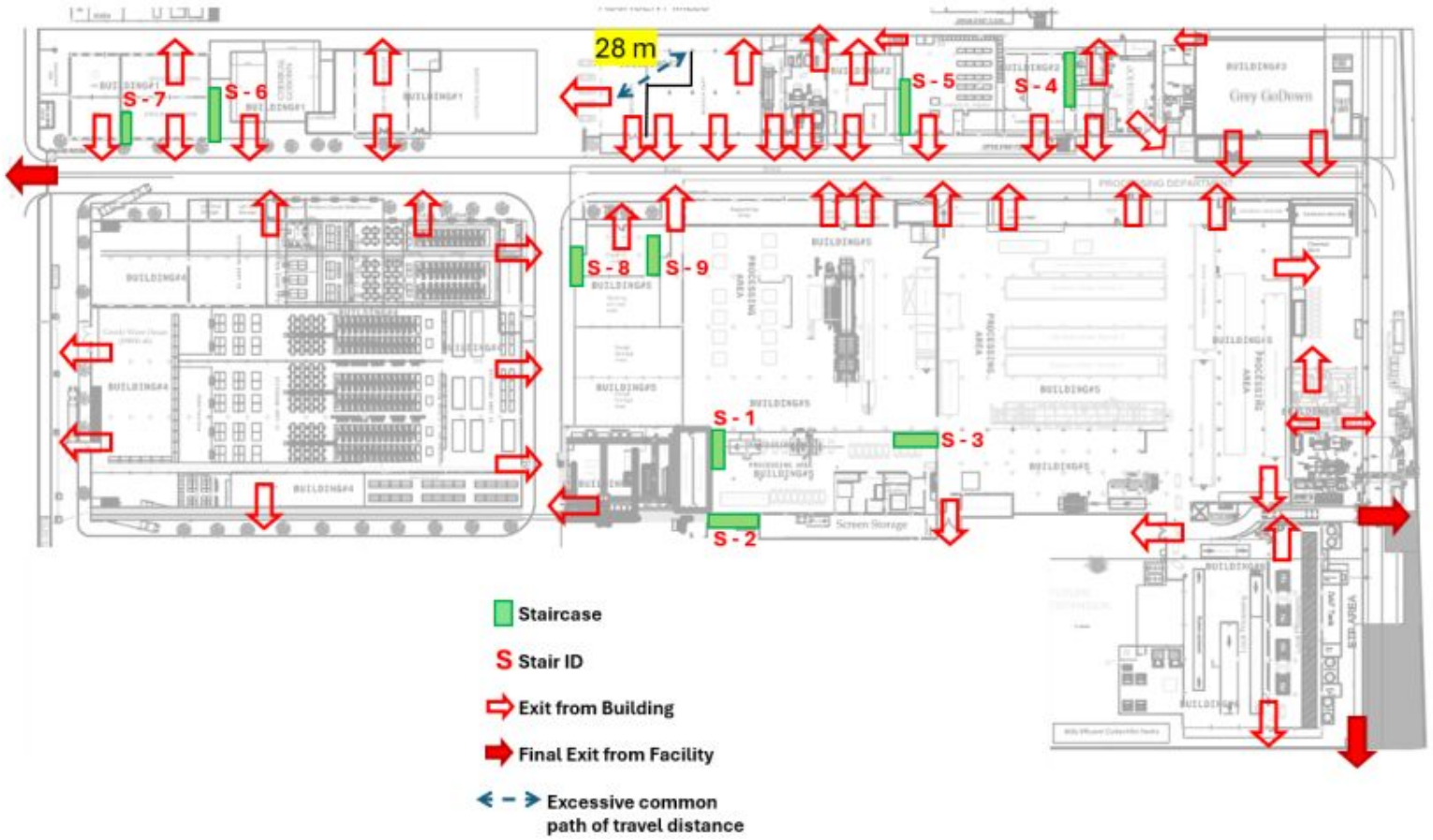


Figure 1.4. Ebrahim Textile Ground Floors Exit plans for complete facility

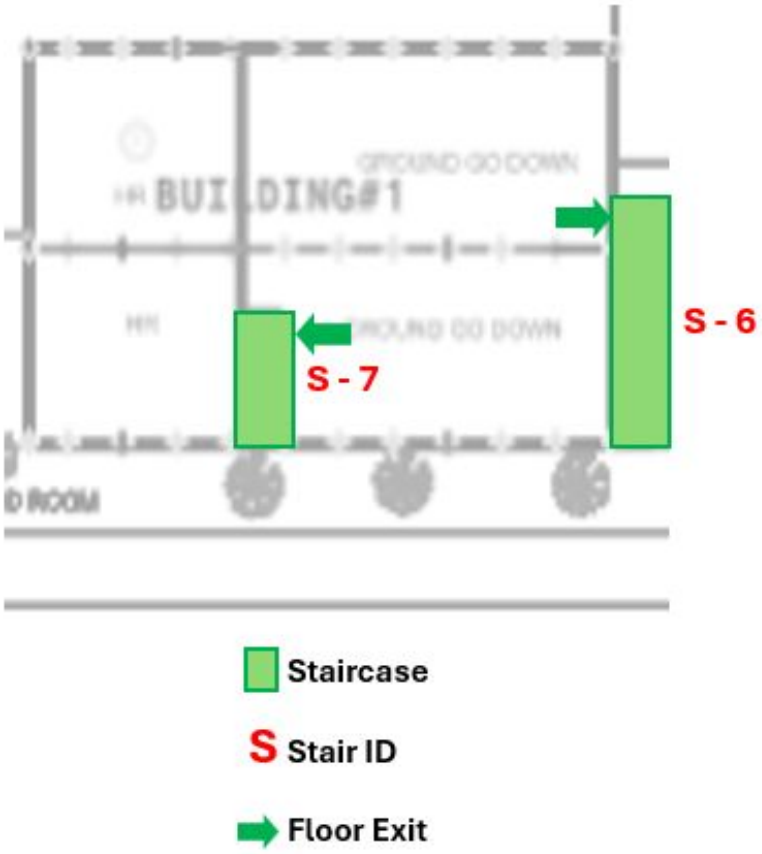


Figure 1.5. Building 1 - 1st Floor Exits Plan



Figure 1.6. Building 2 - 1st Floor Exits Plan



Figure 1.7. Building 5 (Structure 4) - Ground floor Exits Plan

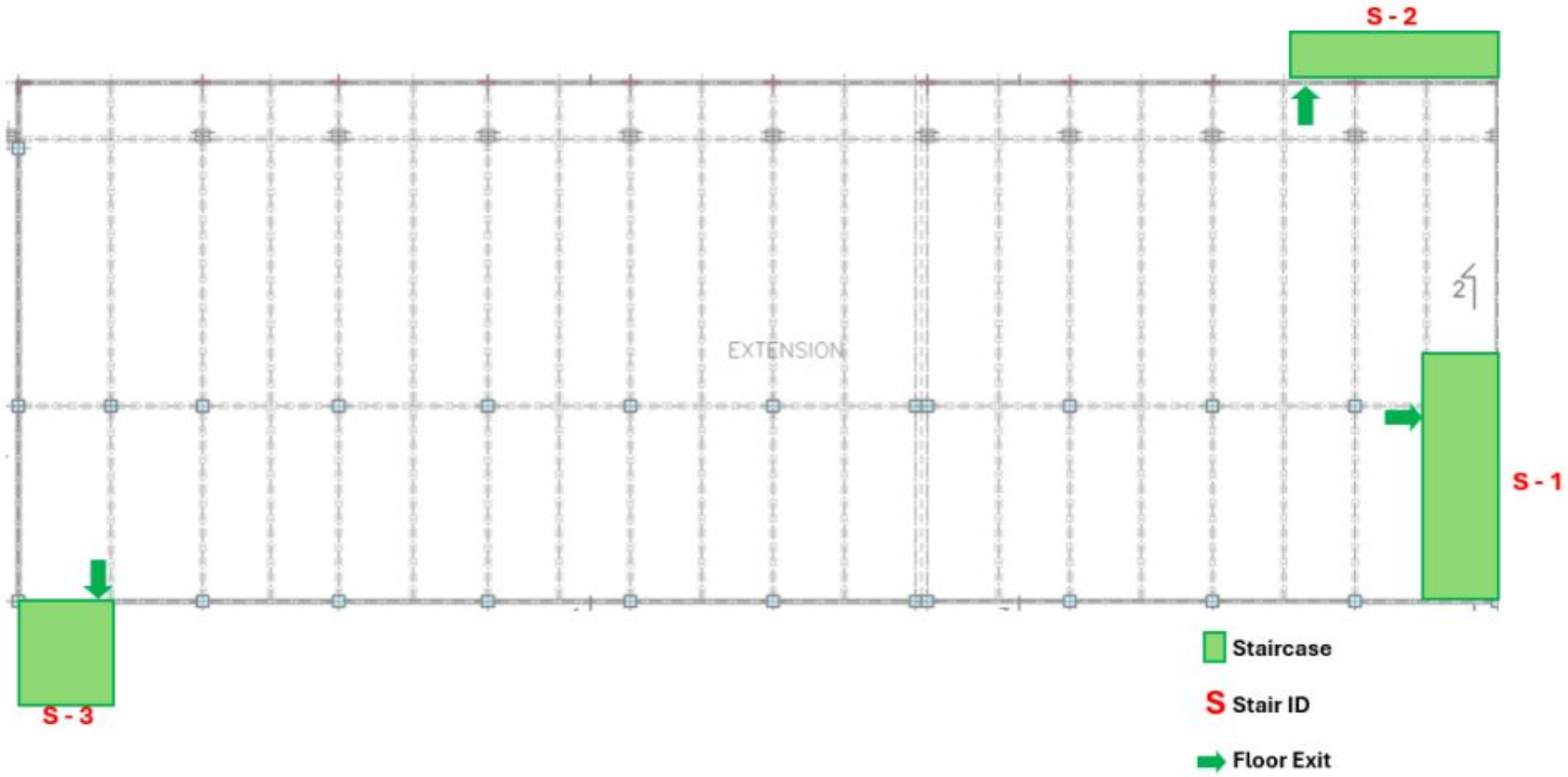


Figure 1.8. Building 5 (Structure 4) - Typical Floor (1st to Roof) Exits Plan

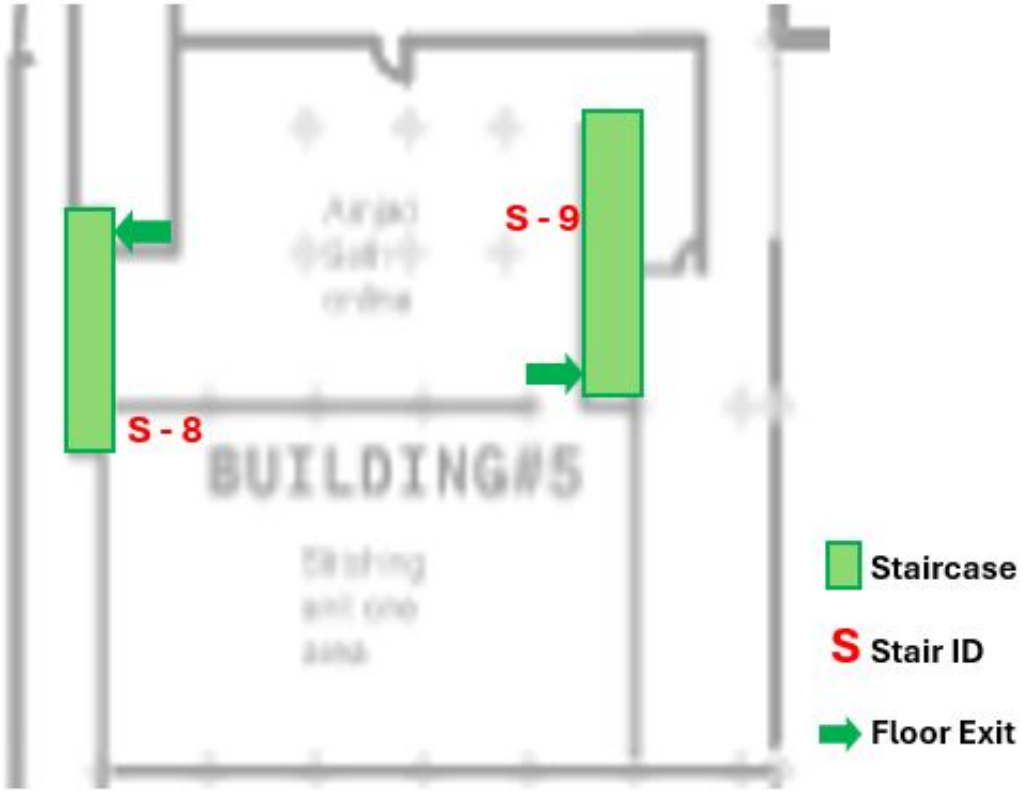


Figure 1.9. Building 5 (Structure 5) - 1st Floor Exits Plan

3.2 Building Use

Floor level	Occupancy	Occupant load	No. of exits	Excessive travel distance		
				Alternatives	Common path	Dead end
Building 1						
Ground	Admin HR, godowns, warehouse, workshop	20	6	-	-	-
1st floor	Empty	-	2	-	-	-
Roof	Unoccupied	-	-	-	-	-
Building 2						
Ground	Digital printing, embroidery, main store, local folding, prayer area, workshops	60	16	-	28 m	-
1st floor	Marketing office	60	2	-	-	-
Roof	Unoccupied	-	-	-	-	-
Building 3						
Ground floor	Greigh department, offices	20	3	-	-	-
Roof	Unoccupied	-	-	-	-	-

Building 4						
Ground	Stitching, warehouse	150	8	-	-	-
Roof	Unoccupied	-	-	-	-	-
Building 5						
Structure 1, 2, 3						
Ground	Processing, folding, storage, washing, labs,	250	15	-	-	-
Roof	Unoccupied	-	-	-	-	-
Structure 4						
Ground	Processing & Folding	20	4	-	-	-
Mezzanine	FGS storage	20	3	-	-	-
1st Floor	Engraving	20	3	-	-	-
2nd Floor	Stitching	20	3	-	-	-
3rd Floor	Stitching	20	3	-	-	-

Roof	Cutting	20	3	-	-	-
Structure 5						
Ground	Cutting	10	2	-	-	-
1st Floor	Quilting	20	2	-	-	-
Roof	Unoccupied	-	-	-	-	-
Building 6						
Ground	Local processing	35	3	-	-	-
Roof	Unoccupied	-	-	-	-	-

There is sufficient exit capacity in the facility however, common path of travel distances issue found during inspection which is mentioned in above table.

4. Documentation

This table shows the details about the documentation provided by the factory.

No.	Description	Availability Yes/No	Created/Updated Date	Status/Remarks
1.	Architectural Layouts	Yes	-	Master plan is available only
2.	Sectional Layouts	Yes	-	-
3.	Fire NOC Certificate	Yes	-	-
4.	Emergency Action Plan	Yes	12/05/2023	Provided but the available data is not in accordance with Pakistan Building Accord standard
5.	Evidence of Fire Drills	Yes	04/02/2025	Provided but the available data is not in accordance with Pakistan Building Accord standard
6.	Maintenance of Fire Pumps	No	-	-
7.	Maintenance of Fire Extinguishers	No	-	-
8.	Maintenance of Standpipe System	No	-	-
9.	Testing of FLS system	No	-	-

10.	Inspection of Fire Hose Reel	Yes	04/04/2025	Provided but the available data is not in accordance with Pakistan Building Accord standard
11.	Inspection of Fire Hydrants	No	-	-
12.	Inspection of Fire Pump	Yes	07/04/2025	Provided but the available data is not in accordance with Pakistan Building Accord standard
13.	Inspection of Fire Extinguishers	Yes	02/04/2025 (Monthly)	Provided but the available data is not in accordance with Pakistan Building Accord standard

Table 1: Documentation

5. Fire Safety Measures

During our fire site inspection of Ebrahim Textile Mills, a number of fire issues were identified.

It is essential to address these fire safety issues promptly to enhance safety and compliance within Ebrahim Textile Mills. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

6. Means of Escape

6.1 Escape doors

Per Pakistan Accord Standard, escape doors must meet several requirements so that they can be relied on for escape, the key requirements include:
- Doors must not have locking devices regardless if the door is kept locked or unlocked (if a lock is provided, e.g. to prevent unauthorized entry, than it must be overridden by panic hardware).
- Doors must have and opening mechanisms that allows for easy opening without key or prior knowledge.
- Doors into stairs, stair discharge doors, final exit doors from site, and all doors serving areas with more than 50 occupants shall swing in the direction of escape
- Sliding doors shall not serve for escape from areas with more than 10 occupants

This is not always achieved, examples include (list not exhaustive):
- Escape doors are lockable at multiple locations e.g. Door at entry to stairs S-1 at 4th floor, door in processing department at ground floor building 5.
- Doors serving areas with over 50 occupants swing against the direction of escape e.g. Office door at 1st floor of building 2.
- Doors into stairs swing against the direction of escape e.g. Door into stairs S-4.
- Sliding / grill doors serve for escape from areas with more than 10 occupants e.g. at ground floor of building 4, sliding door at ground floor of building 2.

Recommended remedial actions are provided below.F-202506-017628

Timeline	Remediation Plan
Immediate FIN-0022539	Remove all locking devices from doors on all egress routes that could prevent opening from inside.
Medium Term FIN-0022540	Escape doors to be side hung doors that open in the direction of escape and provided with easily opening door mechanisms compliant with Accord Building Standard Section 1 Clauses 1.4 and 1.5. Replace sliding doors with swing doors or provide bypass doors nearby or pocket doors built into the existing sliding door, that are side swinging and open in the direction of travel to an exit.

Longer Term FIN-0022541	Monitor escape door to ensure there are no locking devices installed that prevent easy access to emergency exits.



Figure 2.1 - Lockable & grill door at ground floor of building 4

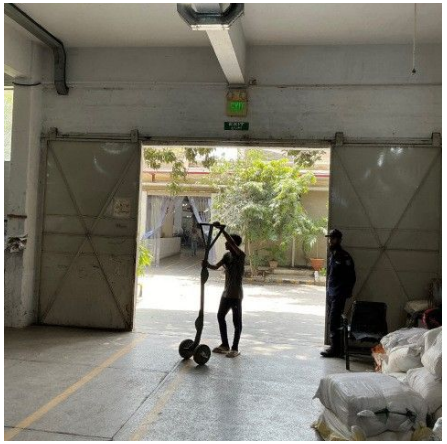


Figure 2.2 - Sliding door at ground floor of building 2



Figure 2.3 - Door into stairs S-4 swing inside at 1st floor



Figure 2.4 - Door at entry to stairs S-1 at 4th floor is lockable

6.2 Exit access

Per Pakistan Accord Standard, escape routes within the building should be sufficiently wide and safe for use in an event of an emergency.

This is not always achieved, examples include (list not exhaustive):
- Obstructions on escape route e.g. Exit access corridor to stairs S-1 have lockable door at 3rd floor of building 5.
- Protected stair discharges inside the building at a distance from building exit e.g. Stairs S-3 discharge inside the building.

Recommended remedial actions are provided below.F-202505-016897

Timeline	Remediation Plan
Immediate FIN-0022524	Remove obstructions on escape routes and keep them clear for escape.
Medium Term FIN-0022525	For stairs discharging within the building, prepare drawings showing the proposed fire separating construction to stair discharge corridor leading outside the building. These shall be submitted to Accord for review and approval prior to the rectification works. The drawings should cover: - Layout showing the location of fire rated walls/doors to protect stair discharge corridors - Details of proposed fire resisting walls and fire door assemblies (e.g. fire door listing, fire rating period, self-closers, door opening direction, details of the door opening mechanism and locks)
Longer Term FIN-0022526	Ensure stair discharge routes / corridors are adequately fire separated from the rest of the building or any fire hazards using 2 hours fire separating construction with 90 minute fire doors. The design shall follow the reviewed proposal and comply with Pakistan Accord Standard. Some of the key requirements include: - All doors into stair discharge corridors need to be fire resisting doors (except stair discharge doors to external air). Penetrations through doors are not permitted. All fire doors require self-closers. Doors into stair discharge routes should be kept closed (but not locked) at all times. - All penetrations through walls into stair discharge corridors need to be fire stopped. Unprotected openings (e.g. holes or window/ louver openings) are not permitted.



Figure 2.5 - Exit access to stairs S-1 have lockable door in the corridor makes obstruction

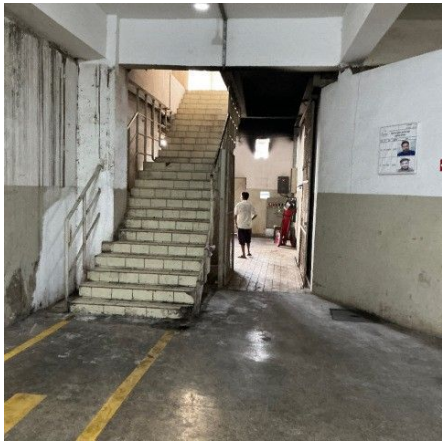


Figure 2.6 - Stairs S-3 discharge inside the building

6.3 Travel distance

Travel distances should meet the limits in the Pakistan Accord Standard. The standard provides limitations on:
- Common path of travel, i.e. the distance the occupants have to travel before there is a choice of escape routes.

This is not always achieved, examples include (list not exhaustive):
- Common path of travel distances exceed permitted limits e.g. Approx. 28m common path has been measured at ground floor in Embroidery department of Building 2.

Recommended remedial actions are provided below.F-202505-016902

Timeline	Remediation Plan
Medium Term FIN-0022533	Propose a plan for addressing the excessive travel distance through alternative escape routes and/or additional exits and submit to Pakistan Accord for review prior to commencing any rectification works on site. Review travel distances from all areas in the building(s) to ensure that they meet the limits in the Pakistan Accord Standard.
Longer Term FIN-0022534	Implement the plan previous reviewed by Pakistan Accord and ensure all travel distances to emergency exits meet the requirements of the Pakistan Accord Standard.



Figure 2.7 - Common path of travel distance has been observed at ground floor of Embroidery department in building 2

6.4 Number of exits

Per Pakistan Accord Standard, the exit provisions should meet the following:
- Minimum 2 exits from every occupied floor.
- Minimum 2 exits from every occupied area with more than 50 occupants.

This is not always achieved, examples include (list not exhaustive):
- There are areas with over 50 occupants that are served by only 1 exit e.g. Prayer area at ground floor, office at 1st floor of building 2.

Recommended remedial actions are provided below.F-202505-016894

Timeline	Remediation Plan
Medium Term FIN-0022514	Review each floor and area to ensure that the number of exits complies with Pakistan Accord Standard. Develop plan and drawings to provide additional exits where necessary. Submit the plan and drawings to Pakistan Accord for review and approval prior to commencing rectification works.
Longer Term FIN-0022515	Implement the above plan as agreed with Pakistan Accord. Number of exits from all floors and parts of the building should comply with Pakistan Accord Standard.



Figure 2.8 - Office at 1st floor of building 2 have single exit to escape

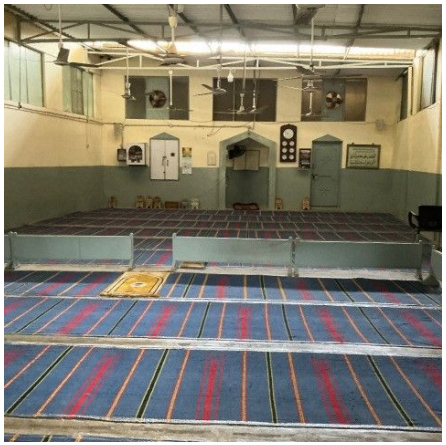


Figure 2.9 - Prayer area at ground floor have single exit to escape

6.5 Protection of stairs

Per Pakistan Accord Standard, exit stairs should be adequately protected, including:
- Stairs to be fire separated from the rest of the building using 2 hour fire resisting construction with 90 minute fire rated doors.
- Stair discharge routes should be protected to the same level of safety as the stair.
- There should be no fire load and/or services passing through a stair.

This is not always achieved, examples include (list not exhaustive):
- Unprotected openings into stairs e.g. at 4th floor of stairs S-1 at 2nd floor of stairs S-3.
- Doors not fire doors e.g. Stairs S-1, S-2 and S-3 have no fire doors at all floors.
- Services passing through or within stairs e.g. at 3rd floor of stairs S-3, 1st floor of stairs S-2.

Recommended remedial actions are provided below.F-202505-016900

Timeline	Remediation Plan
Medium Term FIN-0022531	Prepare drawings showing proposed fire separating construction to protect stairs. These shall be submitted to Pakistan Accord for review and approval prior to rectification works. The drawings should cover: - Layout showing the location of fire rated walls /doors to protect the stairs. - Details of proposed fire resisting walls and fire door assemblies (e.g. fire door listing, fire rating period, self-closers, door opening direction, details of the door opening mechanism and locks). - Location of stair discharge routes and details of fire resisting construction to protect that route.
Longer Term FIN-0022532	Ensure all emergency escape stairs are fire separated from the rest of the building using 2 hours fire separating construction with 90-minute fire doors. The design shall follow Building code of Pakistan Section 10.3.2.1. Key requirements include: - All doors into stairs should be fire resisting doors with self-closing mechanisms. - Penetrations through doors are not permitted. - Doors into stairs should be kept closed (but not locked) at all times.



Figure 2.10 - Services passing through 3rd floor of stairs S-3



Figure 2.12 - Unprotected openings at 2nd floor of stairs S-3



Figure 2.11 - Unprotected opening at 4th floor of stairs S - 1

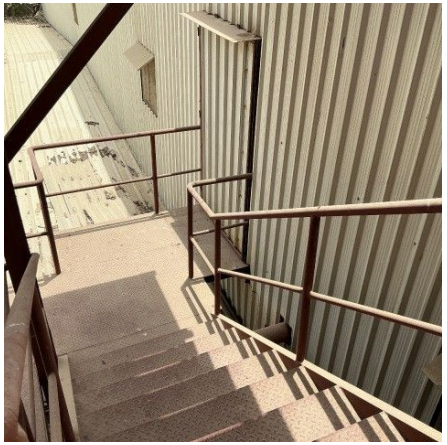


Figure 2.13 - Non rated fire door in stairs S-2 at 2nd floor



Figure 2.14 - Non rated fire door at 3rd floor of stairs S-2



Figure 2.15 - No fire door at 4th floor of stairs S-1

6.6 Escape stair condition

Per Pakistan Accord Standard, escape stairs should be design to be safe for use, including day to day use.

This is not always achieved, examples include (list not exhaustive):
- There are no handrails e.g. Stairs S -1 and S-3 have no handrails on one side of the stairs.
- Stair balustrades are not safe in stairs S-2.

Recommended remedial actions are provided below.F-202505-016899

Timeline	Remediation Plan
Medium Term FIN-0022530	Ensure the following are provided: - Ensure all stairs have balustrades. - Handrails to be provided on both sides of each stair. Ensure the handrails are at appropriate height.



Figure 2.16 - Stairs balustrades in S-2 is not safe

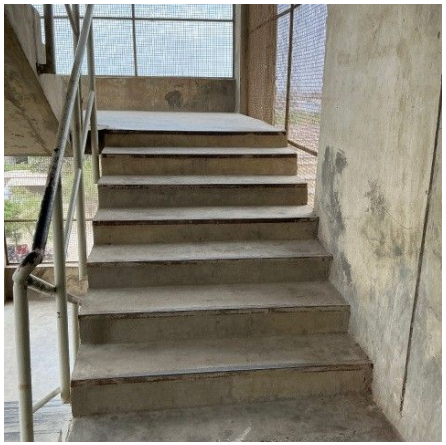


Figure 2.17 - Stairs S-1 have no handrails on one side of the stairs



Figure 2.18 - Stairs S-3 have handrails on single side only

6.7 Exit discharge

Exit discharge routes outside the building should be sufficiently protected so that people can escape safely in the event of fire anywhere in the building. This includes:
- External escape routes should not lead via high risk areas.

This is not always achieved, examples include (list not exhaustive):
- External escape routes are through high risk areas e.g. Exit discharge from building 3 have scrap storage in the route.
- External escape routes are obstructed e.g. Building 2 and Building 4 exit discharge have obstruction.

Recommended remedial actions are provided below.F-202505-016898

Timeline	Remediation Plan
Immediate FIN-0022527	Remove obstructions on external escape routes and keep them clear for escape.
Medium Term FIN-0022528	Prepare drawings showing proposed fire separating construction to exit discharge routes. These shall be submitted to Accord for review and approval prior to commencing rectification works. The drawings should cover: - Layout showing the location of fire rated walls/doors to protect stair discharge routes - Details of proposed fire resisting walls and fire door assemblies required to protect discharge routes
Longer Term FIN-0022529	- Ensure external exit discharge routes are adequately fire separated from the rest of the building. - Ensure external escape routes do not pass close to unprotected high risk areas



Figure 2.19 - Exit discharge from building 4 have obstruction in route



Figure 2.20 - Exit discharge from building 3 have scrap storage



Figure 2.21 - Exit discharge from building 2 have obstructions

6.8 Exit signage

Per Pakistan Accord Standard, exits (e.g. escape stairs or final exits leading directly outside the building) must be clearly marked by approved exit signs. Additionally, escape routes leading to these exits need to be signed where the escape route is not readily apparent. The aim of these provisions is for the occupants to be able to easily find and escape route in an event of an emergency.

The following deficiencies have been observed on site, including (list not exhaustive):
- Missing exit signs in corridors and circulation areas e.g. at 4th floor and 3rd floor of building 5
- Missing exit signs in production areas e.g. at ground floor of building 2 and building 3. at 3rd floor of building 5.
- Missing exit signs in offices and back of house areas e.g. at 1st floor of building 2.
- Missing exit signs at exits from site
- Missing exit signs at roof
- Exit signs are not clear / misleading e.g. at 1st floor of building 5 production area, exit signs into stairs S-1 at 4th floor.
- Electrically powered exit signs indicate battery problems

Recommended remedial actions are provided below.F-202505-016896

Timeline	Remediation Plan
Immediate FIN-0022521	Remove misleading directional escape signs.
Medium Term FIN-0022522	Rectify and/or provide exit signage so access to exits is marked by approved, readily visible signs from all areas of the floor and clearly visible at exits, to comply with Pakistan Accord Standard and BCP 10.15.1, 10.15.2 and 10.15.3.
Longer Term FIN-0022523	Ensure that all exit signage is checked periodically and in good working condition.



Figure 2.22 - Exit signs are missing at 1st floor of building 2



Figure 2.24 - Misleading exit signs at 1st floor of building 5



Figure 2.23 - Exit signs are missing at 3rd floor of building 5



Figure 2.25 - Exits signs are missing at ground floor of building 2



Figure 2.26 - Misleading signs at entry to S-1 stairs at 4th floor of building 5

7. Fire Rated Construction

7.1 Separation of high-risk areas

Per Pakistan Accord Standard, increased fire risk areas require to be fire separated from the rest of the building using the following:
- Electrical rooms, Generator rooms, Fire Pumps room, and Hazardous waste room: 2 hour fire resisting construction with 90 minutes fire doors

The following deficiencies have been observed on site, including (list not exhaustive):
- Fire pump room is not fire separated from the rest of the building
- Electrical rooms are not fire separated from rest of the building e.g. at ground floor of building 5, Electrical DB is installed at 1st floor of stairs S-5 entry
- Generator room is not fire separated
- Boiler area is not fire separated e.g. at ground floor of building 5.
- Mechanical plant areas are not fire separated e.g. Compressors and storage is not fire separated at ground & 1st floor of building 5.

Recommended remedial actions are provided below.F-202505-016890

Timeline	Remediation Plan
Medium Term FIN-0022501	Prepare drawings showing the proposed fire separation details of increased fire hazard areas. These shall be submitted to Accord for review and approval prior to the rectification works. The drawings should cover: - Locations of all increased fire hazard areas in the building (e.g storage, electrical rooms, mechanical rooms, fire pump room etc.) - Locations of fire rated walls and doors to protect these rooms - Details of proposed fire rated walls and fire doors (e.g. door listing, self-closers, door opening direction)
Longer Term FIN-0022502	Provide 2-hour fire rated construction including 90-minute fire rated doors assemblies according to the layouts and documents reviewed by Pakistan Accord.



Figure 2.27 - Electrical room is not fire separated from generator at ground floor of building 5



Figure 2.29 - Electrical DB is installed at 1st floor of stairs S-5 entry makes the escape is passing through high hazard area



Figure 2.28 - Boiler is not fire separated at ground floor of building 5



Figure 2.30 - Compressors are not fire separated from production area at ground floor of building 5

7.2 Separation of shafts

Per Pakistan Accord Standard, shafts should be enclosed in 2 hour fire resisting construction to maintain compartmentation between floors.

The following deficiencies have been observed on site, including (list not exhaustive):

- Lift shafts are not separated with fire rated construction e.g. lift shaft in building 5.

Recommended remedial actions are provided below.F-202507-019335

Timeline	Remediation Plan
Medium Term FIN-0025248	Prepare drawings showing the proposed fire separation details of vertical shafts. These shall be submitted to Accord for review and approval prior to the rectification works. The drawings should cover: - Locations of fire rated walls and doors enclosing vertical shafts. - Details of fire rated walls and fire doors (e.g. listing, fire rating period, self-closers, door opening direction)
Longer Term FIN-0025249	Provide 2-hour fire rated protection to lift shafts according to the layouts and documents reviewed by Pakistan Accord.



Figure 2.31 - Lift shaft in building 5 is not fire separated in all floors

7.3 Floor openings

Per Pakistan Accord Standard, each floor should be fire separated using 2 hours fire resisting construction with 90 minutes fire rated door/shutter assemblies.

The following deficiencies have been observed on site, including (list not exhaustive):
- Open voids connecting different levels e.g. at 3rd floor of building 5, 1st to 2nd floor of building 5
- Chutes and/or conveyor belts are not fire separated e.g. from 4th floor till 1st floor in building 5.

Recommended remedial actions are provided below.F-202506-017629

Timeline	Remediation Plan
Medium Term FIN-0022542	Prepare a plan and drawings showing the proposed fire separation details. These shall be submitted to Accord for review and approval prior to the rectification works. The drawings should cover: - Locations of floor openings - Proposed fire separation details (including fire rating of walls and doors)
Longer Term FIN-0022543	Provide fire rated separation between floors by implementing the plan reviewed previously by Pakistan Accord.



Figure 2.32 - Floor opening at 3rd floor in building 5



Figure 2.33 - Conveyor chute from 4th floor to 1st floor in building 5



Figure 2.34 - Floor openings from 1st floor to 2nd floor in building 5

7.4 Separation of occupancies

Pakistan Accord Standard requires different occupancy types to be fire separated from each other. The aim is to limit the fire to only one occupancy type.

The following deficiencies have been observed on site, including (list not exhaustive):
- Prayer area with over 50 occupants is not fire separated from the rest of the building at ground floor of the building.

Recommended remedial actions are provided below.F-202505-016889

Timeline	Remediation Plan
Medium Term FIN-0022499	Prepare a plan and drawings showing showing the proposed separation details. These shall be submitted to Pakistan Accord for review and approval prior to the rectification works. The drawings should cover: - Different occupancy types in the building. - Proposed fire separation details (including fire rating of walls and doors).
Longer Term FIN-0022500	Provide fire rated separation between occupancies as per proposal reviewed by Pakistan Accord.



Figure 2.35 - Storage & Prayer area is not fire separated at ground floor

7.5 Storage

Per Pakistan Accord Standard, extensive storage areas need to be fire separated from the rest of the building using 2 hour fire resisting walls with 90 minutes fire rated doors.

Limited in-process storage is permitted in production areas, subject to meeting all of the following criteria:
- Combined storage areas do not exceed 10% of the production floor area
- Storage height is less than 2.45m
- Storage areas are grouped into islands not exceeding 23 sq.m.
- Islands are separated from other storage islands by at least 3m gap.

The following deficiencies have been observed on site, including (list not exhaustive):
- Storage areas are not fire separated from the rest of the building e.g. Storage and cutting area is not fire separated at 5th floor of building 5, ground of Building 5, ground floor of building 4.
- Presence of areas where storage does not meet miscellaneous in-process storage requirements (listed above) e.g. Excessive storage is found at 4th floor of Building 5, 3rd floor of building 5.

Recommended remedial actions are provided below.F-202505-016888

Timeline	Remediation Plan
Medium Term FIN-0022497	Identify all designated storage areas. Identify which storage areas can be re-arranged to meet the in-process storage limitations described above and which storage areas need will be fire separated from the rest of the building. For all storage areas which require fire separation, prepare a drawing presenting the proposed fire separating construction details and submit to Pakistan Accord for review prior to the rectification works. The drawings should cover: - Locations of fire rated walls and doors - Details of fire rated walls and fire doors (e.g. listing, fire rating period, self-closers, door opening direction)
Longer Term FIN-0022498	Ensure that the above plan is implemented on site following the previously reviewed documentation and in accordance with Pakistan Accord Building Standard.



Figure 2.36 - Excessive storage at 4th floor of building 5



Figure 2.38 - Storage is not fire separated from production at ground floor of Building 5



Figure 2.37 - Storage & Cutting area is not fire separated at 5th floor of building 5

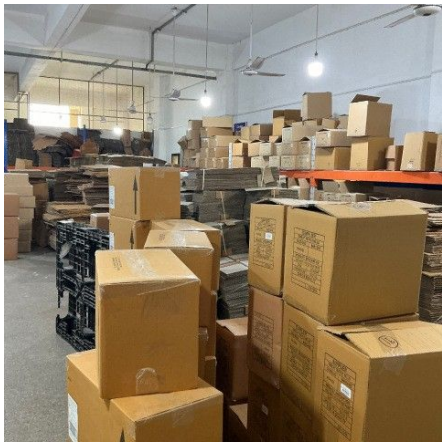


Figure 2.39 - Excessive storage at 3rd floor of building 5

8. Fire Suppression

8.1 Sprinkler system

Per Pakistan Accord Standard, a fire sprinkler system is required in buildings exceeding 18.3 m in height when measured from the fire department vehicle access level to the highest occupied floor. The system should be designed and installed in accordance with NFPA 13.

The following deficiencies have been observed on site, including (list not exhaustive):
- Sprinkler system is not provided in building 5 (all structures) as the height of this building is 20.25m which is exceeds the threshold of 18.3m for sprinklers system installation.

Recommended remedial actions are provided below.F-202506-017820

Timeline	Remediation Plan
Medium Term FIN-0023031	(1) Submit design documentation for a sprinkler system (along with required standpipe system) and supporting infrastructure to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets and product certification of all listed equipment to show full compliance of the Suppression System (SUPS) with NFPA 13, NFPA 14, 20, 22, and 24. (2) Submit documentation to Pakistan Accord showing that an L/C has been opened for import of any components required for the SUPS design submitted earlier.

Longer Term FIN-0023032	(3) Submit documentation to Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on site for installation of the SUPS. (4) Once installation of the SUPS system is complete, factory to submit full ""as-built"" documentation of the system to Pakistan Accord for review. (5) Factory to advise Pakistan Accord when SUPS installation is fully completed and fully functional. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation. (6) Provide full testing and commissioning (T&C) certification from the SUPS system installer and make provision for attendance of the system installer, factory technical personnel and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the SUPS system will be complete.

8.2 Standpipe system

Per Pakistan Accord Standard a Class-I standpipe systems is required in the building. The system should be designed and installed in accordance with NFPA 14.

The following deficiencies have been observed on site, including (list not exhaustive):
- Connection points are not compliant at building 5 stairs.
- Missing standpipe outlets in stairs at 4th floor of building 5.

Recommended remedial actions are provided below.F-202505-016895

Timeline	Remediation Plan
Immediate FIN-0022518	Replace any damaged equipment and ensure current system is fully operational.
Medium Term FIN-0022519	Provide the design for a Class 1 standpipe system with adequate risers and connections, in accordance with NFPA 14 as part of the supporting infrastructure to the SUPS system, and submit to Accord for review
Longer Term FIN-0022520	Provide a Class 1 standpipe system with adequate risers and connections, in accordance with NFPA 14 as part of the supporting infrastructure to the SUPS system.



Figure 2.40 - Standpipe outlets are missing in building 5 at 4th floor

8.3 Fire pumps

Per Pakistan Accord Standard, a complete fire pump set is required. The pump set should be designed and installed in accordance with NFPA 20. The system should comprise of: a) Electric pump (Main), b) Diesel Pump (Back-Up), and c) Jockey Pump. Each pump should be provided with their individual drivers and controllers. Pumps shall be installed with positive suction. These measures are intended to provide a level of resilience whereby a failure of a pump or lack of power in the building will not compromise fire water supply in the building.

The following deficiencies have been observed on site, including (list not exhaustive):

- Fire pump set is not listed.

Recommended remedial actions are provided below.F-202507-019336

Timeline	Remediation Plan
Medium Term FIN-0025250	Provide design for electric duty pump, diesel backup fire pumps and jockey pump, together with individual drivers and controllers, in accordance with NFPA 20 as part of the supporting infrastructure to the SUPS system and submit to Pakistan Accord for review.
Longer Term FIN-0025251	Provide electric duty pump, diesel backup fire pumps and jockey pump, together with individual drivers and controllers, in accordance with NFPA 20 as part of the supporting infrastructure to the SUPS system.



Figure 2.41 - Fire pump set is non listed installed in the factory

9. Hazardous Materials

9.1 Chemicals storage

This relates to storage, handling, and use of flammable chemicals and combustible liquids and solids.
Chemical storage areas should be fire separated from the rest of building. Adequate ventilation should be provided.

The following deficiencies have been observed on site, including (list not exhaustive):
- Flammable materials are not stored in a fire separated room
- Inadequate ventilation of chemical storage areas

Recommended remedial actions are provided below.F-202506-017630

Timeline	Remediation Plan
Medium Term FIN-0022547	Factory to collect MSDS sheets for all chemicals stored, classify materials, and propose storage arrangements to meet NFPA 30 requirements. Prepare drawings showing the proposed chemical storage areas. These shall be submitted to Accord for review and approval prior to the rectification works. The drawings should cover: - Locations of all chemical storage areas - Locations of fire rated walls and doors to protect these rooms - Details of proposed fire rated walls and fire doors (e.g. door listing, self-closers, door opening direction) - Ventilation provisions to each of these rooms
Longer Term FIN-0022548	Implement the above plan on site. All chemical storage areas to be compliant with Pakistan Accord Standard.



Figure 2.42 - Chemical store at ground floor is not fire separated in building 1

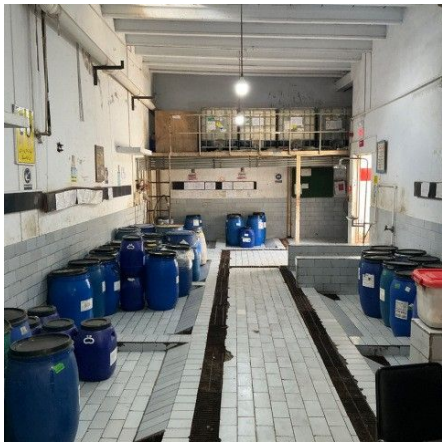


Figure 2.43 - Chemical store at ground floor is not fire separated in building 5

9.2 Gas storage

Flammable gas storage areas should be fire separated from the rest of building. Adequate ventilation should be provided.

The following deficiencies have been observed on site, including (list not exhaustive):
- There is no automatic gas shut off valve in main gas supply line.

Recommended remedial actions are provided below.F-202507-019337

Timeline	Remediation Plan
Medium Term FIN-0025262	Prepare drawings showing the location and installation of automatic gas shut off valve in the main supply line. These shall be submitted to Accord for review and approval prior to the rectification works.
Longer Term FIN-0025263	Implement the above plan on site. All gas storage areas to be compliant with Pakistan Accord Standard.

10. Fire Safety Management

10.1 Fire drills

Per Pakistan Accord Standard, fire drills shall be conducted as a minimum twice a year. The aim of these drills is to make sure that all staff is adequately trained and suitable provisions are in place to enable timely evacuation in the event of an emergency.

The following deficiencies have been observed on site, including (list not exhaustive):
- Fire drill documentation is of poor standard
- Not all staff are involved in fire drills

Recommended remedial actions are provided below.F-202506-017627

Timeline	Remediation Plan
Medium Term FIN-0022535	Conduct a fire drill and make records, including findings and recommended future improvements.
Longer Term FIN-0022536	Ensure fire drills are performed a minimum 2 times per year.

10.2 Records of system testing

Per Pakistan Accord Standard, all fire safety systems should be adequately maintained to ensure they will operate effectively in an event of an emergency.
All fire and life safety systems should be maintained in accordance with different standards applicable depending on the system type, e.g. fire suppression systems to NFPA 25, fire alarm and detection system to NFPA 72, fire extinguishers to NFPA 10 etc.

The following deficiencies have been observed on site, including (list not exhaustive):
- Inadequate records of maintenance of Fire Pumps
- Inadequate records of maintenance of Standpipes
- Inadequate records of maintenance of Fire Extinguishers

Recommended remedial actions are provided below.F-202505-016893

Timeline	Remediation Plan
Medium Term FIN-0022512	- Prepare a detailed schedule of proposed system maintenance. Identify who is responsible for filling in the details. - Identify how systems will be maintained and who is responsible. - The schedule to be submitted to Accord for review and approval.
Longer Term FIN-0022513	- Ensure all systems are being maintained as per the schedule agreed with Accord . - Keep maintenance records.

10.3 Records of systems maintenance

Per Pakistan Accord Standard, all fire safety systems should be adequately maintained to ensure they will operate effectively in an event of an emergency.
All fire and life safety systems should be maintained in accordance with different standards applicable depending on the system type, e.g. fire suppression systems to NFPA 25, fire alarm and detection system to NFPA 72, fire extinguishers to NFPA 10 etc.

The following deficiencies have been observed on site, including (list not exhaustive):
- Inadequate records of maintenance of Fire Pumps
- Inadequate records of maintenance of Standpipes
- Inadequate records of maintenance of Fire Extinguishers

Recommended remedial actions are provided below.F-202505-016892

Timeline	Remediation Plan
Medium Term FIN-0022509	- Prepare a detailed schedule of proposed system maintenance. Identify who is responsible for filling in the details. - Identify how systems will be maintained and who is responsible. - The schedule to be submitted to Accord for review and approval.
Longer Term FIN-0022510	- Ensure all systems are being maintained as per the schedule agreed with Accord . - Keep maintenance records.

10.4 Emergency Action Plan

Per Pakistan Accord Standard, and Emergency Action Plan shall be developed for each building. The plan shall follow the guidance in NFPA 2800 “Standard on Facility Emergency Action Plans” 2023.
Evacuation plans shall be posted at each exit stair and final exit to the outside. Evacuation plans should indicate the location of the plan (i.e. you are here sign), the location of nearby exits, manual call points, and first-aid fire fighting provisions.
An Emergency Action Plan (EAP) report is required for the site. EAP should describe in detail what should happen in the event of a fire, this includes the roles and responsibilities, training needs, emergency equipment requirements, communication protocols etc. EAP should be practiced as part of evacuation drills.

The following deficiencies have been observed on site, including (list not exhaustive):

- There is EAP document which lacks the details procedures, roles and responsibilities in the event of fire.

Recommended remedial actions are provided below.F-202505-016891

Timeline	Remediation Plan
Medium Term FIN-0022503	Ensure that evacuation plans are provided on each level. Prepare an Emergency Action Plan report.
Longer Term FIN-0022504	Ensure emergency evacuation plans are provided and up to date. Ensure EAP is being practiced as part of fire drills.



Figure 2.44 - EAP document is not in accordance with Pakistan Accord Building Standard

11. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.